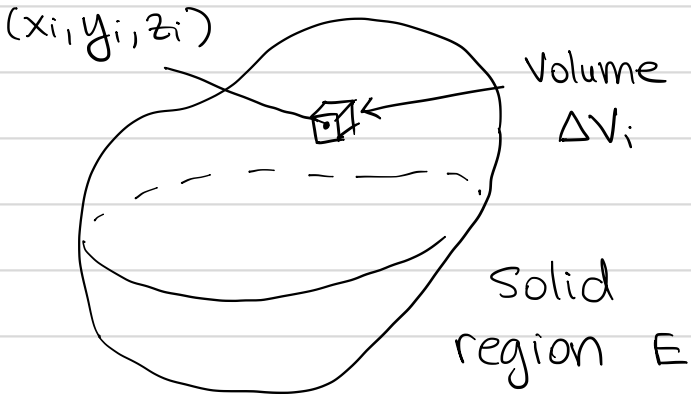
Lecture 5

Triple integrals (15.6)



Definition:

$$\iiint_E f(x,y,z)\,dV = \lim_{\Delta V_i \to 0} \underbrace{\sum_i f(x_i,y_i,z_i)\Delta V_i}_{\text{Riemann sum}}$$

Fubini: $E = [a,b]\times[c,d]\times[r,s]$

$$\iiint_E f(x,y,z)\,dV =$$

$$= \int_r^s \int_c^d \int_a^b f(x,y,z)\,\underbrace{dx\,dy\,dz}_{\text{in any of the 6 possible orders}}$$

Example: $B = [0,1]\times[-1,2]\times[0,3]$

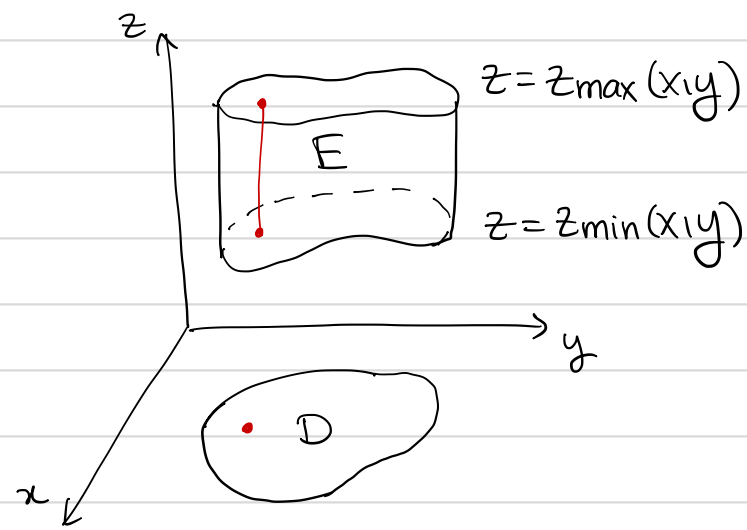
$$f(x,y,z) = xyz^2$$

$$\iiint_B f\,dV = \int_0^1\int_{-1}^2\int_0^3 xyz^2\,dz\,dy\,dx$$

$$= \int_0^1\int_{-1}^2 xy\,\frac{z^3}{3}\Big|_0^3\,dy\,dx$$

$$= 9\int_0^1 x\left[\frac{y^2}{2}\right]_{y=-1}^{2} dx$$

$$= \frac{27}{2}\int_0^1 x\,dx = \frac{27}{4} \;//$$



$$\boxed{\iiint_E f\,dV = \iint_D \left[\int_{z_{min}(x,y)}^{z_{max}(x,y)} f(x,y,z)\,dz\right] dA}$$

E is a solid region of <u>type 1</u>

There exist also

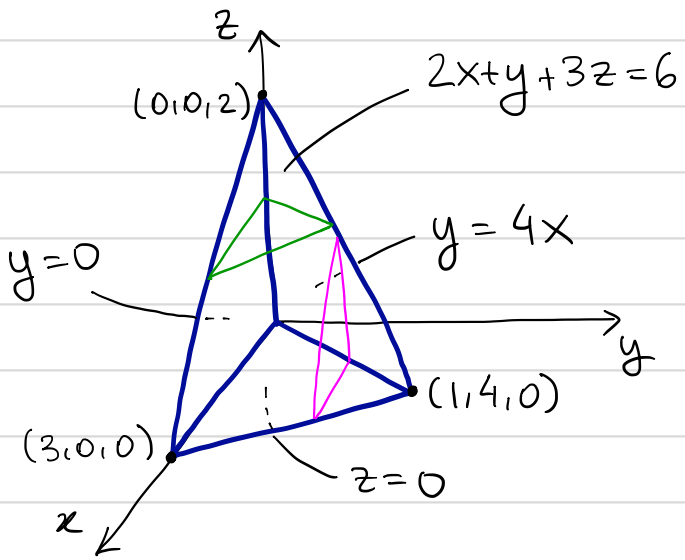
- type 2: D in yz-plane
- " 3: " " xz-plane

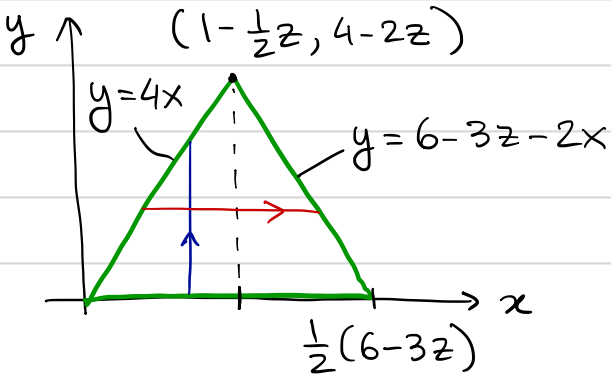
If D is itself type 1, then:

$$\iiint_E f(x,y,z)\,dV =$$

$$= \int_{x_{min}}^{x_{max}} \int_{y_{min}(x)}^{y_{max}(x)} \int_{z_{min}(x,y)}^{z_{max}(x,y)} f(x,y,z)\,dz\,dy\,dx$$

<u>Example</u>: E = tetrahedron



$$\iiint_E f(x,y,z)\,dV =$$

$$= \int_{z=0}^{2} \int_{y=0}^{4-2z} \int_{x=\frac{1}{4}y}^{\frac{1}{2}(6-y-3z)} f\,dx\,dy\,dz$$